FPGA 8x1 Multiplexer – Hardware Test Results

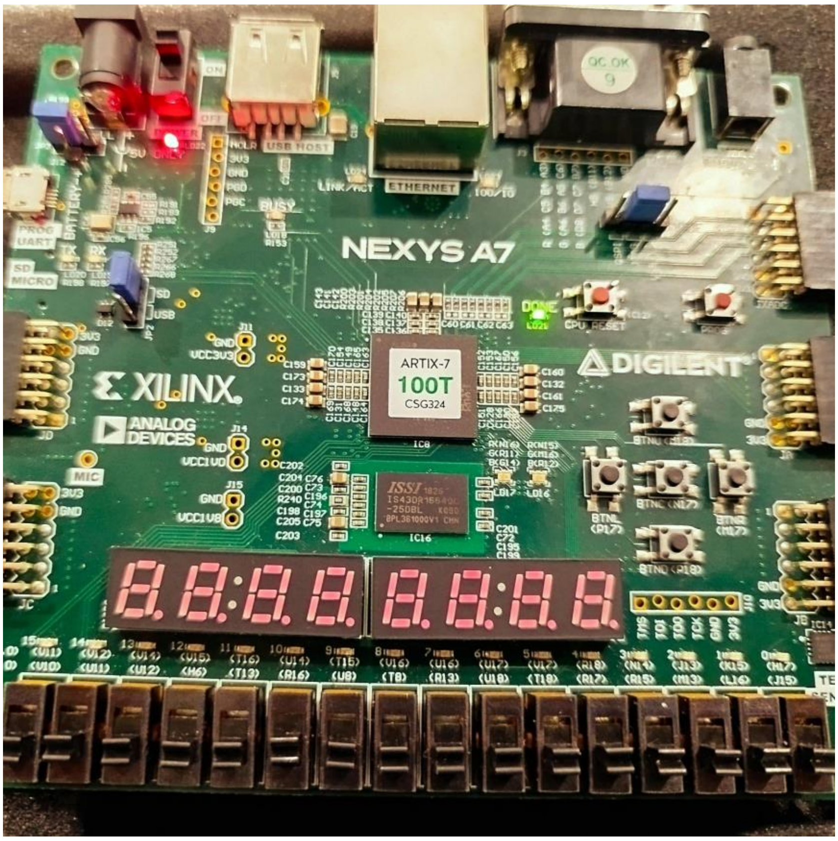
This document presents hardware test results of an 8x1 multiplexer implemented on an FPGA board. The following photographs demonstrate correct operation of the design under different input and selection conditions.

LED Mapping Information

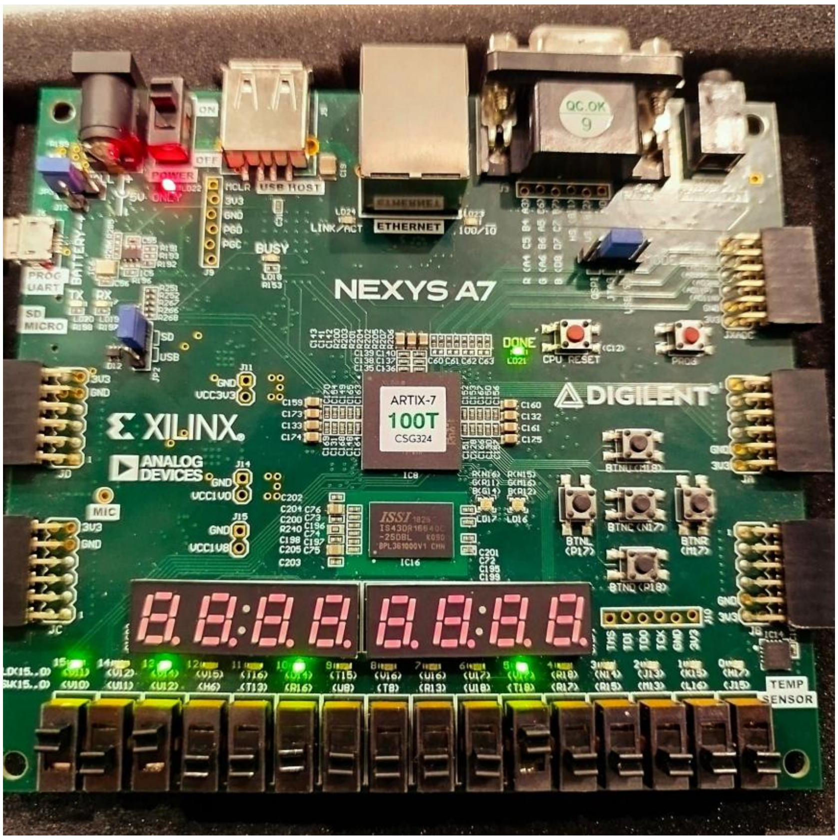
In this project, the LEDs on the FPGA board are used to visualize the multiplexer signals.

- LEDs 0–7 show the states of the MUX inputs.
- LED 10 shows the MUX output (Y).
- LEDs 13, 14, and 15 represent the selection inputs, displaying the currently selected input in binary form.

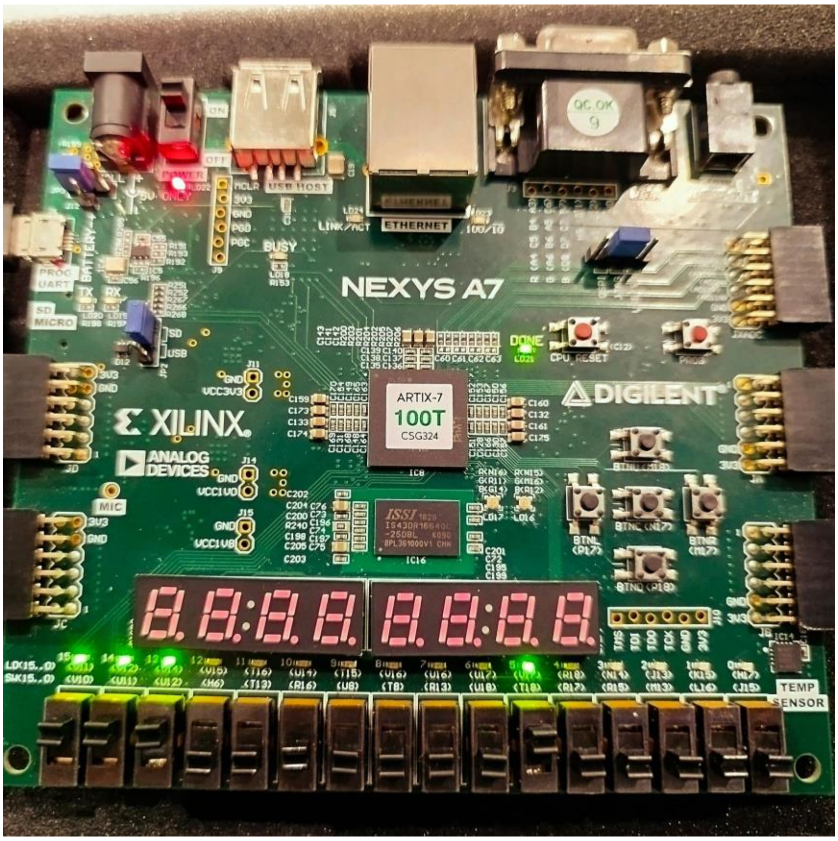
Test 1: Selection bits = 000, Inputs = 00000000, Output = 0



Test 2: Selection bits = 101, Inputs = 00100000 (I_5 active), Output = 1



Test 3: Selection bits = 111, Inputs = 00100000 (I_5 active), Output = 0



Test 4: Selection bits = 111, Inputs = 10100000 (I_7 and I_5 active), Output = 1